WEEK 2 · LESSON 7 OF 24 · HANDOUT

Data: Features, Labels and Quality

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=0yCJMt9Mx9c — open the lesson online to watch inside the course.

🎓 **Go deeper:** Stanford Machine Learning, Lecture 1 — Andrew Ng — https://www.youtube.com/watch?v=UzxYlbK2c7E

Your goal in this lesson: Explain what features and labels are, describe the main types of data, and spot common data-quality problems before they ruin a model.

Data is the fuel

A machine-learning model can only be as good as the data it learns from. Experienced teams often spend **most of a project's time on data** — finding it, cleaning it and checking it — and much less on the model itself. A popular saying sums it up: **garbage in, garbage out.**

Rows, features and labels

Most learning data can be pictured as a table. Imagine a cooperative that wants to predict whether a member will repay a small loan:

Member	Age	Monthly sales (RWF)	Years in cooperative	Past late payments	Repaid?
A	34	180,000	5	0	**Yes**
B	27	60,000	1	3	**No**
C	45	250,000	9	1	**Yes**

- Each **row** is one example (one member).
- Each input column is a **feature** — a measurable property used to make the prediction.
- The column we want to predict is the **label** (also called the **target**): *Repaid?*

A model learns how the features relate to the label, then predicts the label for new members.

Types of data

- **Numerical** — numbers you can do maths with: age, sales, temperature.
- **Categorical** — a choice from a list: district, crop type, phone brand.
- **Text** — messages, reviews, documents.
- **Images, audio and video** — photos of leaves, voice notes, CCTV.
- **Time series** — values in time order: daily rainfall, hourly electricity use.

Tables with clear columns are called **structured** data. Text, images and sound are **unstructured** — richer, but harder to use. Deep learning is especially good with unstructured data.

Common data-quality problems

Problem	Example	Why it hurts
Missing values	Sales left empty for 20% of members	The model learns from an incomplete picture
Errors and typos	Age recorded as 340	Strange values distort the patterns
Duplicates	The same member entered twice	Some examples count double
Inconsistent formats	"Kigali", "kigali", "KGL"	One place looks like three
Wrong labels	A repaid loan marked "No"	The model learns the wrong lesson
Not representative	Data only from urban members	The model fails for rural members
Out of date	Prices from five years ago	The world has changed

The last two are the most dangerous because they are invisible in a quick look. A dataset can be perfectly clean and still **not represent the people the model will serve**.

Data leakage — a hidden trap

Leakage happens when a feature secretly contains the answer. If a column says "sent to debt collection", it already reveals the loan was not repaid. The model will look brilliant in testing and fail in real life, because that information does not exist at the moment of the real decision.

Key terms

- **Feature** — an input property used for prediction.
- **Label / target** — the answer the model should predict.
- **Dataset** — the full collection of examples.
- **Data cleaning** — fixing or removing errors, gaps and duplicates.
- **Representative data** — data that matches the people and situations the model will meet.

Remember this: Features are inputs, the label is the answer, and data quality — especially being representative and free of leakage — matters more than any clever model.

Name: _____ Date: _____

Activity

Task 1

Design a small dataset (10 rows) for a problem in your community: for example, predicting if a student will finish secondary school, or a shop's daily sales.

Task 2

Name at least four features and the label.

Task 3

Write down three data-quality problems your data could have, and how you would fix each.

Task 4

Check for leakage: is any feature only known *after* the answer?

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. In a loan-repayment dataset, what is the label?
   - A. The table's title
   - B. The member's name
   - C. Whether the loan was repaid
   - D. The member's age
2. What is a feature?
   - A. A measurable input property used to make a prediction
   - B. A bug in the code
   - C. The final answer
   - D. The name of the model
3. Photos, voice notes and documents are examples of…
   - A. Duplicates
   - B. Labels
   - C. Structured data
   - D. Unstructured data
4. Data collected only from urban members is used for a national model. What is the main problem?
   - A. There are too many columns
   - B. The data is numerical
   - C. The data is too clean
   - D. The data is not representative
5. A feature 'sent to debt collection' is used to predict repayment. This is an example of…
   - A. Good feature engineering
   - B. Missing values
   - C. Data leakage
   - D. Clustering
6. What does 'garbage in, garbage out' mean?
   - A. A model trained on poor data gives poor results
   - B. Computers produce waste
   - C. Delete old files often
   - D. Clean data is not needed

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